

MARCO SOBRIDO ALVANO

Computer Engineering Student · Full-Stack Developer · Buenos Aires, Argentina

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EDUCATION

Universidad Argentina de la Empresa (UADE) — B.S. in Computer Engineering (Ingeniería en Informática) 2025 – 2030 (*expected*)

Relevant coursework: Algorithm Design (Programming III), Data Structures & Algorithms II, Object-Oriented Programming (Java), Operating Systems, Databases (SQL Server), Information Systems I & II, Requirements Engineering, Calculus II.

TECHNICAL EXPERIENCE

Independent Full-Stack Developer — self-directed agtech platform, built solo 2025 – Present

Next.js · React · TypeScript · Supabase (Postgres, Auth, RLS) · Web Bluetooth · Playwright · Docker · Google Cloud

- Designed, built and shipped a **multi-tenant web platform** end to end: workspace-scoped roles (admin / user / read-only) enforced in the database with **Postgres Row-Level Security**, invite codes and Google sign-in.
- Integrated handheld **RFID readers over Web Bluetooth** (BLE, Nordic UART); found that browsers cannot reach Bluetooth Classic (SPP) readers, which shaped the architecture: a PWA over BLE plus a native-Android path for SPP.
- Made it **offline-first**: edits queue locally and sync with retries and an audit trail. Wrote regression tests after tracing silently lost writes to supabase-js returning, rather than throwing, server rejections.
- Used **Playwright** to automate submissions to a government web portal with no public API; containerized the service with **Docker** (health checks) and deployed it to **Google Cloud Run** via Cloud Build and Artifact Registry.
- Added a strict Content-Security-Policy, formula-injection-safe CSV/Excel exports and WCAG AA design tokens.

HACKATHONS

Platanus Hack 26 — Vibefence · AI Security track (team) Buenos Aires · May 2026

36-hour hackathon, 100+ hackers · Python · FastAPI · Next.js · Supabase · Playwright · MCP

- With my team, built a **runtime security layer for AI coding agents** (Claude Code, Cursor): a local agent intercepts every tool call and blocks shell commands whose authority traces back to low-trust sources, such as a prompt-injected README.
- The build also included an agentic red-team that verifies access-control bugs (IDOR, broken auth) with reproducible evidence, one-click rollback of destructive migrations via parallel Postgres schemas, and a live dashboard on Vercel.

Serious Game Hackathon: Tourism — Madrid in Game × BIG Buenos Aires (team of 5) Jul 2026

El Ateneo: Cuentos del Splendid · JavaScript (HTML5 Canvas, WebAudio) · Unreal Engine 5

- Our team designed a cozy management game set in El Ateneo Grand Splendid, a 1919 theater turned bookstore, and shipped a **playable 3-level browser prototype** (sorting, café service, upgrades, NPC stories from verified history) plus a pitch deck.
- Built a **1:1-scale recreation of the building in Unreal Engine 5** as an explorable level with custom materials, working from researched architectural references (≈20 m dome, horseshoe balconies).

PROJECTS

E-commerce Storefronts — apparel shop and hunting & sport-shooting retailer 2026

Next.js 16 · React 19 · TypeScript · Supabase (Postgres, RLS) · Vercel

- Checkout **recalculates every price on the server** from the database (clients only send items and quantities), and RLS makes orders insert-only, so no customer can read another's orders.
- Built versioned SQL migrations, tests that run against the live database, a full-purchase SQL test wrapped in a rollback, per-product legal sale rules and daily exchange-rate pricing in ARS.

Graph Algorithms REST API — metro fiber-optic network (Programming III, team of 3) Aug 2026 – Present

Java · Spring Boot · Neo4j · JUnit · Docker Compose

- Spring Boot API over a 10-node Neo4j fiber network exposing **BFS/DFS, Dijkstra, Floyd-Warshall, Prim/Kruskal (union-find), greedy, 0/1-knapsack DP, backtracking and branch & bound**, with hand-verified JUnit tests.

SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL

Web & Backend: React, Next.js (App Router), Node.js, REST APIs, Spring Boot, Tailwind CSS, PWAs, Web Bluetooth

Data & Cloud: PostgreSQL, Supabase (Auth, RLS), Neo4j, SQL Server, Google Cloud (Cloud Run, Cloud Build), Vercel, Docker

Tools: Git & GitHub, Linux (Red Hat), Playwright, Figma, Unreal Engine 5, AI-assisted development (Claude Code)

CERTIFICATIONS & LANGUAGES

Red Hat System Administration I (RH124), 2026 · **SAP ERP Fundamentals** (LinkedIn Learning), 2026 · **Cambridge B2 First**, 2024

Languages: Spanish (native), English (B2, Cambridge)